

Kevin Xiang Li

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SKILLS

Artificial Intelligence: Agent eval and data; Post-training with SFT and RL

Machine Learning: SGLang, vLLM, Unsloth, JAX, Nsight, Gymnasium, HuggingFace, PyTorch, Python

Mobile Dev: Flutter, SQLite, Rust **AR/VR:** Unreal, Unity, C# **Web:** TypeScript, JavaScript, HTML, CSS

EDUCATION

- **Stanford University** Stanford, CA, U.S.A
M.S. in Computer Science; GPA: 3.9 2024. 9 – 2026. 6
- **University of Michigan** Ann Arbor, MI, U.S.A
B.S. in Computer Science, Minor in Linguistics; GPA: 3.8, Summa Cum Laude. 2020. 9 – 2024. 5

EXPERIENCES

- **Harbor: Open Evaluator for Agents** Remote
Core Contributor, advised by Ludwig Schmidt, under review at NeurIPS 2026 2026. 2 - Present
 - **Co-led Harbor-Index curation and qualitative analysis:** Organized a team of 14 to audit 300 challenging tasks across 40 benchmarks to arrive at a high-quality, challenging, and diverse subset as Harbor-Index; Designed analysis pipeline to detect and fix task issues such as wrong gold answer, broken verifier, and reward hacks.
 - **Expanded Harbor's benchmark coverage to novel tasks:** Added MedAgentBench adapter (first agentic benchmark for healthcare environments); Added CooperBench adapter (first multi-agent coding benchmark). Contributed adapters served as exemplars for new benchmark authors.
 - **Hardened SWE-bench adapter against test poisoning:** Fixed a harness vulnerability that let agent patches override held-out tests, prevented reward hacking in Harbor and helped fix SWE-bench upstream.
 - **Improved adapter contribution flow:** Refactored and updated `harbor adapters init` to jump-start contribution; Streamlined bulky parity result upload with a new agent skill, saving 10+ hours of git pull time and 60+GB/1M+ files worth of redundant storage per adapter author.
- **SWE-chat: Coding Agent Interactions From Real Users in the Wild** Stanford University
Researcher at Stanford, advised by Diyi Yang and Sanmi Koyejo, under review at COLM 2026 2026. 2 – 6
 - **Curated large-scale human-AI interaction data:** Collected SWE-chat, the first large-scale dataset of real-world coding agent trajectories, covering 6K sessions, 56K user prompts, and 330K tool calls.
 - **Engineered automated data pipelines:** Built scalable systems to continuously extract, parse, and structure raw agent session logs from public repositories, achieving precise line-level code attribution between humans and AI.
 - **Designed rigorous annotation codebooks:** Developed and validated annotation guidelines (achieving high inter-annotator agreement) to classify user intents and agent failure at scale with LLM-as-a-judge.
 - **Uncovered real-world AI usage patterns:** Revealed code comprehension (18.9%) outpaces generation as the top intent. Vibe coding sessions doubled in 3 months to account for 37% of sessions, yet human-agent collaborative sessions are 2.6x more time efficient and 3x more token efficient.
 - **Quantified agent failure & inefficiency:** Found less than 44% of agent code survives into commits, with autonomous coding doubling per-line costs. Exposed a severe oversight imbalance: agents can operate 100+ minutes autonomously but request clarification in just 1.4% of turns, triggering user pushback 50% of the turns.
- **Marin: An Open Lab for Building Foundation Models** Stanford University
Researcher at Stanford, advised by David Hall and Percy Liang 2025. 9 - 2026. 6
 - **Released the largest open collection of SWE traces with 112B tokens:** Covered 12.3M rollouts, 122K PRs, 3K repos, and 16 languages; Trended #2 globally on Hugging Face datasets; Validated data quality by fine-tuning Qwen3-1.7B-Base, reaching 11% on SWE-bench Verified up from 0%, with consistent improvement verified up to 7.5 B training tokens.

- **Reproduced NemotronTerminal on Terminal-Bench 2:** Set up Marin SFT and eval to reproduce NemotronTerminal-8B with TPU v5p-32 on 366K samples. Training took 5 days and resulted in +15.9% on TB2.
 - **Reproduced RL results on key reasoning benchmarks:** Aligning in-house RL framework to match baseline on MATH-500, fixed training dataset, prompt format, reward function, advantage function, and loss.
 - **Scaled training throughput:** Achieved a 3.7x speedup in async RL trainer throughput (from 4.1 to 15.4 requests/s) by implementing batch-dynamic max sequence lengths in the trainer.
 - **Open source contribution to vLLM:** Identified and resolved a critical bug in the vLLM-tpu sampling logic where default top K values incorrectly forced greedy sampling regardless of temperature settings.
- **SWE-smith Multilingual: Scaling Multilingual Data for SWE Agents** Stanford University
Researcher at Stanford, advised by John Yang 2025. 9 - 2026. 3
- **Released largest dataset of software engineering tasks:** Synthesized 386K bug-fix pairs spanning 39 repos.
 - **Boosted issue resolve rate from 0% to 2.3%:** Fine-tuned Qwen2.5-Coder-32B using 142 GLM-4.6 teacher trajectories, improving SWE-bench Rust resolve rate from 0% to 2.3% and increasing completion rate by 2.4x.
 - **Slashed task synthesis time from days to hours:** Built a scalable pipeline using Modal to parallelize task synthesis across hundreds of concurrent sandboxes, reducing processing time from days to hours.
 - **Designed procedural modifications for 7 languages:** Added 70+ procedural modifications for efficient coding task synthesis across C++, Rust, TypeScript, JavaScript, Java, Ruby, PHP
 - **Contributed 19 PRs to SWE ecosystem:** Maintaining SWE-smith (14 PRs merged); 2 pending at SWE-bench; 1 merged/1 pending at SWE-ReX, 1 merged into mini-swe-agent.
- **SGLang: Benchmarking and Accelerating Vision-Language Models** Santa Clara, CA
ML Engineer at Nvidia 2025. 6 - 2025. 9
- **Benchmarked VLMs on large GPU clusters:** Measured throughput and latency of the Qwen 2.5 VL family (3B–72B) across H200 and B200 clusters.
 - **Pinpointed and reported inference bottlenecks:** Used Nsight, NVTX markers, and PyTorch Profiler to pinpoint kernel-level bottlenecks in SGLang and vLLM; dissected framework performance gaps under varying concurrency and provided detailed reports well received by both SGLang and vLLM multimodal teams.
 - **Submitted 5 PRs to SGLang that boosted Qwen 2.5 VL throughput by 1.6x end-to-end on MMMU:** (1) Doubled Qwen 2.5 VL vision prefill speed via automatic attention backend selection, (2) Accelerated rotary embedding with CUDA rotary kernels, boosting vision prefill throughput by 21%, (3) Identified and removed redundant device-to-host visual feature transfers to enable accelerated GPU hashing, yielding 7.5% end-to-end speedup on MMMU, (4) Fused SwiGLU in ViT to double peak TensorCore utilization, resulting in 4.5% vision prefill throughput gain, (5) Unified VLM benchmarking to support reliable cross-framework comparisons.
- **VideoMultiAgents: A Multi-Agent Framework for Video QA** Stanford University
Researcher, in collaboration with Panasonic, advised by Ehsan Adeli 2024. 10 - 2025. 3
- **Designed multi-agent framework with modality-specific agents for video QA:** Enhanced video understanding by leveraging strengths of video, text, and graph modalities through multi-agent collaboration.
 - **Discovered that modality-specific multi-agent architectures benefit from structure and independence:** Showed that our Report architecture performs the best by aggregating opinions from independent modality-specific agents through an organizer agent and weighing strength of evidence from each modality.
 - **Achieved SOTA accuracy on popular video QA benchmarks:** Improved previous SOTA on Intent-QA by +6.2%, EgoSchema subset +3.4%, and NExT-QA by +0.4%.
- **MVUBench: First Contamination-Free Code Completion Benchmark** University of Michigan
Researcher, advised by Cyrus Omar 2023. 9 - 2024. 8
- **Built MVUBench for repo-scale code completion evaluation:** Built a contamination-free benchmark of unit-tested Model-View-Update (MVU) web apps for repo-scale evaluation of code completion and retrieval.
 - **Grounded LLMs with static retrieval:** Retrieved type definitions and function headers from language servers to ground LLMs in repo context, leading to 3.5x higher resolve rate than vector retrieval with GPT-4.
 - **Powered test-time scaling with static error feedback:** Developed an agent harness that raised GPT-4 resolve rate by up to 5x through iterative repair of syntax and type errors reported by the language server.
 - **Published at OOPSLA 2024:** Second-author publication at top programming languages conference.